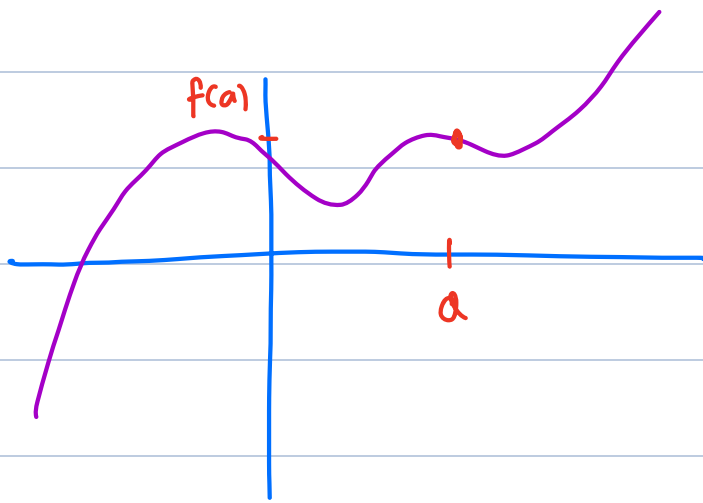


Calc 1 Lecture 4 : Evaluating limits

Simplest case: for "nice" functions $f(x)$
as long as a is in the domain of $f(x)$,
then

$$\lim_{x \rightarrow a} f(x) = f(a)$$



Definition: A function $f(x)$ is continuous at
a point a if $\lim_{x \rightarrow a} f(x) = f(a)$

intuitively: a function is continuous if you can draw
it without lifting the pen (no sudden jumps)

Theorem: the following families of functions are all continuous whenever they are defined.

- Polynomials, rational functions, root functions
- trig functions, inverse trig functions
- exponentials, logarithms

Continuity laws

Theorem If f and g are continuous at a , then so are the following functions:

1. $f + g$
 2. $f - g$
 3. cf
 4. fg
 5. f/g
- } still continuous
(where c is a constant)
(if $g(a) \neq 0$)

Theorem "composition of continuous functions is continuous"

If f is continuous at a and g is continuous at $\underline{f(a)}$ then $g \circ f$ is continuous at a

Bottom line: any function you can write down in terms of functions in our library of functions is continuous wherever it's defined (exception: piecewise functions)

In this case you can evaluate a limit

by direct substitution,

$$\lim_{x \rightarrow a} f(x) = f(a)$$

if f cont.
at a

Example a) Evaluate $\lim_{x \rightarrow \pi} \frac{\sin(x)}{2 + \cos(x)}$

b) Evaluate $\lim_{x \rightarrow 1} \frac{\sqrt{x^3 + 2}}{\ln(2x)}$

a) check the function is continuous:

- $\sin(x)$ is continuous
- $\cos(x)$ is continuous
- 2 is continuous (a constant function)

$\Rightarrow 2 + \cos(x)$ still continuous (continuity law)

$\Rightarrow f(x) = \frac{\sin(x)}{2 + \cos(x)}$ still continuous wherever it's defined (continuity law)

check the function is defined at $x = \pi$

$$f(\pi) = \frac{\sin(\pi)}{2 + \cos(\pi)} = \frac{0}{2 + (-1)} = \frac{0}{1} = 0.$$

$\Rightarrow$ So we can evaluate by direct substitution

$$\lim_{x \rightarrow \pi} \frac{\sin(x)}{2 + \cos(x)} = \lim_{x \rightarrow \pi} f(x) = f(\pi) = \boxed{0}$$

direct substitution

b) $\lim_{x \rightarrow 1} \frac{\sqrt{x^3 + 2}}{\ln(2x)}$

check continuous:

- $x^3 + 2$ continuous (polynomial)
- $\sqrt{x}$ continuous (root function)
- by composition, $\sqrt{x^3 + 2}$ continuous
- $\ln(2x)$ continuous
- by division, $f(x) = \frac{\sqrt{x^3 + 2}}{\ln(2x)}$ continuous (whenever it's defined)

check defined at $x=1$: $f(1) = \frac{\sqrt{1^3 + 2}}{\ln(2)} = \frac{\sqrt{3}}{\ln(2)}$

$$\lim_{x \rightarrow 1} \frac{\sqrt{x^3 + 2}}{\ln(2x)} = \frac{\sqrt{1^3 + 2}}{\ln(2)} = \boxed{\frac{\sqrt{3}}{2}}$$

direct
substitution.
 $x=1$

To summarize:

Limit Technique #1 (direct substitution)

if $f(x)$ is continuous at $x=a$

then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Limits by simplification

Example: Find $\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$

observation: direct substitution doesn't work.

plug in $x=1$: $\frac{1^2 - 1}{1 - 1} = \frac{0}{0}$ undefined.

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x+1)(\cancel{x-1})}{\cancel{x-1}} = \lim_{x \rightarrow 1} x+1$$

$$= 1+1 = \boxed{2}$$

direct subst.
 $x=1$

Careful: $\frac{x^2 - 1}{x - 1} = \frac{(x+1)(\cancel{x-1})}{\cancel{x-1}} = x+1$

undefined when $x=1$ $\nearrow$ $\boxed{x-1}$ $\nearrow$ $\cancel{x-1}$ $\uparrow$ defined when $x=1$

is incorrect because the two functions have different domains

Limit Technique #2 (removable discontinuity)

given a function $f(x)$ which is not defined at $x=a$
try to find a "simpler" function $g(x)$ which is
defined at $x=a$. (often by cancellation), such that
 $g(x) = f(x)$ when $x \neq a \Rightarrow \lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$

More examples Find the following limits

$$a) \lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$$

$$b) \lim_{x \rightarrow -1} \frac{x^2 - 2x - 3}{x - 3}$$

$$c) \lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2}$$

$$d) \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1}$$

$$e) \lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x - 3}$$

Solutions:

$$a) \lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3} = \lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$$

$$= \lim_{x \rightarrow 3} x+1 = 3+1 = \boxed{4}$$

direct
subst.
 $x=3$

$$b) \lim_{x \rightarrow -1} \frac{x^2 - 2x - 3}{x - 3} = \lim_{x \rightarrow -1} \frac{(x-3)(x+1)}{x-3}$$

$$= \lim_{x \rightarrow -1} x+1 = -1+1 = \boxed{0}$$

OR in this case the function is already
defined at $x = -1$

can do direct subst. immediately

$$c) \lim_{x \rightarrow 2} \frac{x^3 - 8}{x - 2} = \lim_{x \rightarrow 2} \frac{x^3 - 2^3}{x - 2}$$

"differences of cubes formula"

$$a^3 - b^3 = (\underline{a} - \underline{b})(a^2 + ab + b^2)$$

$$\underline{x^3 - 2^3} = (x-2)(x^2 + 2x + 4)$$

$$\lim_{x \rightarrow 2} \frac{x^3 - 2^3}{x-2} = \lim_{x \rightarrow 2} \frac{\cancel{x-2} (x^2 + 2x + 4)}{\cancel{x-2}}$$

$$= \lim_{x \rightarrow 2} x^2 + 2x + 4$$

$$= \begin{array}{l} \text{direct} \\ \text{subst.} \\ x = 2 \end{array} \quad 2^2 + 2 \cdot 2 + 4 = 4 + 4 + 4 = \boxed{12}$$

$$d) \lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1}$$

two approaches:

Approach 1: $x - 1 = (\sqrt{x})^2 - 1^2$
 $= (\sqrt{x} - 1)(\sqrt{x} + 1)$

$$\frac{\sqrt{x} - 1}{x - 1} = \frac{\cancel{\sqrt{x} - 1}}{(\cancel{\sqrt{x} - 1})(\sqrt{x} + 1)}$$

Approach 2: Idea: whenever you see a $\sqrt{\quad}$ consider the option of "multiplying and dividing by conjugate"

$$a + \sqrt{b} \quad \xRightarrow{\text{conjugate}} \quad a - \sqrt{b}$$

$$(a + \sqrt{b}) \cdot (a - \sqrt{b}) = a^2 - (\sqrt{b})^2 = a^2 - b$$

$$\frac{\sqrt{x} - 1}{x - 1} = \frac{\sqrt{x} - 1}{x - 1} \cdot \frac{\overbrace{\sqrt{x} + 1}^{\text{conjugate}}}{\sqrt{x} + 1}$$

$$= \frac{(\sqrt{x})^2 - 1^2}{(x - 1)(\sqrt{x} + 1)} = \frac{x - 1}{(x - 1)(\sqrt{x} + 1)}$$

$$\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{\cancel{x - 1}}{(\cancel{x - 1})(\sqrt{x} + 1)} = \lim_{x \rightarrow 1} \frac{1}{\sqrt{x} + 1}$$

direct subst.
 $x = 1$

$$\frac{1}{\sqrt{1} + 1} = \boxed{\frac{1}{2}}$$

$$e) \lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x-3} \quad \text{"0/0"}$$

conjugate of $\sqrt{x+1} - 2$ is $\sqrt{x+1} + 2$

multiply top and bottom by conjugate:

$$\lim_{x \rightarrow 3} \frac{\sqrt{x+1} - 2}{x-3} \sim \lim_{x \rightarrow 3} \frac{(\sqrt{x+1} - 2)(\sqrt{x+1} + 2)}{(x-3)(\sqrt{x+1} + 2)}$$

$$= \lim_{x \rightarrow 3} \frac{(\sqrt{x+1})^2 - 2^2}{(x-3)(\sqrt{x+1} + 2)} = \lim_{x \rightarrow 3} \frac{\cancel{x-3} - 4}{(\cancel{x-3})(\sqrt{x+1} + 2)}$$

$$= \lim_{x \rightarrow 3} \frac{1}{\sqrt{x+1} + 2} = \frac{1}{\sqrt{3+1} + 2} = \frac{1}{2+2} = \boxed{\frac{1}{4}}$$

direct
subst.
 $x=3$

More examples:

$$\lim_{x \rightarrow 1} \frac{x^2}{x-1}$$

here simplification doesn't work.

notice denominator $(x-1) \rightarrow 0$ as $x \rightarrow 1$

numerator $x^2 \rightarrow 1$ as $x \rightarrow 1$

intuitively " $\frac{1}{0}$ " approach an infinite limit.

More on infinite limits later...

Limit Laws

Suppose that $\overbrace{\lim_{x \rightarrow a} f(x)}^{L_1}$ and $\overbrace{\lim_{x \rightarrow a} g(x)}^{L_2}$ exist. Then:

$$1. \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = L_1 + L_2$$

$$2. \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = L_1 - L_2$$

$$3. \lim_{x \rightarrow a} [c f(x)] = c \lim_{x \rightarrow a} f(x) = c L \quad (\text{where } c \text{ is some constant})$$

$$4. \lim_{x \rightarrow a} [f(x) g(x)] = \left(\lim_{x \rightarrow a} f(x) \right) \left(\lim_{x \rightarrow a} g(x) \right) = L_1 \cdot L_2$$

$$5. \lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L_1}{L_2} \quad (\text{as long as } L_2 \neq 0)$$

*note: these laws only make sense if the limits L_1 and L_2 are finite

Example: Suppose you know that $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$
and $\lim_{x \rightarrow 0} (1+x)^{1/x} = e$

Then evaluate:

a) $\lim_{x \rightarrow 0} \frac{x}{\sin(x)}$ b) $\lim_{x \rightarrow 0} (1+x)^{2/x}$ c) $\lim_{x \rightarrow 0} \left(\frac{2 \sin(x)}{x} - (1+x)^{1/x} \right)$

a) $\lim_{x \rightarrow 0} \frac{x}{\sin x} = \lim_{x \rightarrow 0} \frac{1}{\left(\frac{\sin x}{x} \right)} \stackrel{\text{limit law 5}}{=} \frac{\lim_{x \rightarrow 0} 1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} \left. \begin{array}{l} \} 1 \\ \} 1 \end{array} \right\} = \frac{1}{1} = \boxed{1}$

b) $\lim_{x \rightarrow 0} (1+x)^{2/x} = \lim_{x \rightarrow 0} \left((1+x)^{\frac{1}{x} + \frac{1}{x}} \right)$
 $= \lim_{x \rightarrow 0} \left((1+x)^{\frac{1}{x}} \cdot (1+x)^{\frac{1}{x}} \right) \stackrel{\text{limit law 4}}{\downarrow}$
 $= \underbrace{\lim_{x \rightarrow 0} \left((1+x)^{\frac{1}{x}} \right)}_e \cdot \underbrace{\lim_{x \rightarrow 0} \left((1+x)^{1/x} \right)}_e$
 $= e^2$

Super limit law: "continuous functions preserve limits"

If $\lim_{x \rightarrow a} f(x) = L$

if $g(x)$ is a function which is continuous at L

then:

$$\lim_{x \rightarrow a} g(f(x)) = g\left(\lim_{x \rightarrow a} f(x)\right) = g(L)$$

Example: Find $\lim_{x \rightarrow 1} \sqrt{\frac{x^2-1}{x-1}}$

use $g(x) = \sqrt{x}$
can move outside
limit sign
note $\sqrt{x}$ is continuous

$$= \sqrt{\lim_{x \rightarrow 1} \left(\frac{x^2-1}{x-1} \right)}$$

earlier: found $\lim_{x \rightarrow 1} \frac{x^2-1}{x-1} = 2$

$$= \boxed{\sqrt{2}}$$

Example: Evaluate $\lim_{x \rightarrow 0^+} \ln(\sin x) - \ln(x)$

solution: first use log properties

$$\ln(\sin x) - \ln(x) = \ln\left(\frac{\sin x}{x}\right)$$

know: $\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$

• $\ln$ is continuous

$$\bullet \ln\left(\lim_{x \rightarrow 0^+} \frac{\sin x}{x}\right) = \ln 1 = 0$$

cont.uous
functions
preserve limits

$$\lim_{x \rightarrow 0^+} \ln\left(\frac{\sin x}{x}\right) = 0$$

* note: all aforementioned limit laws are valid for one-sided limits as well

(can replace $\lim_{x \rightarrow a}$ with $\lim_{x \rightarrow a^+}$ or $\lim_{x \rightarrow a^-}$)

Limit technique #3: Using limit laws

use limit laws 1-5 above, or "super limit law" (continuous functions preserve limits) to infer new limits from old ones

Limits of piecewise functions

Example: a) If $f(x) = \begin{cases} \sqrt{x-4} & \text{if } x > 4 \\ 8-2x & \text{if } x < 4 \end{cases}$

Evaluate $\lim_{x \rightarrow 4} f(x)$ (if it exists)

Idea: evaluate both one-sided limits and compare

$$\bullet \lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4^+} \sqrt{x-4} = \sqrt{4-4} = \sqrt{0} = 0$$

$\uparrow$
can assume $x > 4$ direct substitution $x = 4$

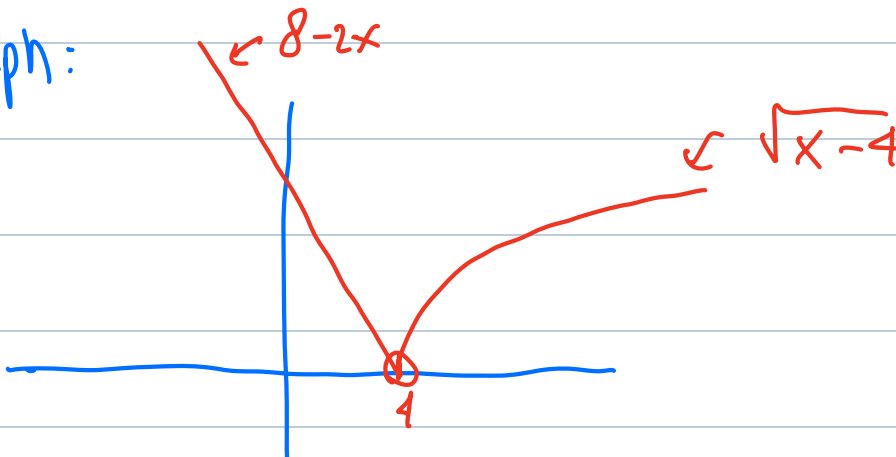
$$\bullet \lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} (8-2x) = 8-2 \cdot 4 = 0$$

$\uparrow$
can assume $x < 4$ direct subst. $x = 4$

in conclusion: $\lim_{x \rightarrow 4^+} f(x) = 0 = \lim_{x \rightarrow 4^-} f(x)$

$$\Rightarrow \boxed{\lim_{x \rightarrow 4} f(x) = 0}$$

graph:



b) If $f(x) = \frac{|x-2|}{x^2-4}$, evaluate $\lim_{x \rightarrow 2} f(x)$ (if it exists)

Hint: it helps to write $|x-2|$ as a piecewise function

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x \leq 0 \end{cases}$$

$$|x-2| = \begin{cases} x-2 & \text{if } x-2 \geq 0 \\ -(x-2) & \text{if } x-2 \leq 0 \end{cases}$$

$$|x-2| = \begin{cases} x-2 & \text{if } x \geq 2 \\ 2-x & \text{if } x \leq 2 \end{cases}$$

$$\frac{|x-2|}{x^2-4} = \begin{cases} \frac{x-2}{x^2-4} & \text{if } x \geq 2 \\ \frac{2-x}{x^2-4} & \text{if } x \leq 2 \end{cases}$$

take both one-sided limits

$$\bullet \lim_{x \rightarrow 2^+} \left(\frac{|x-2|}{x^2-4} \right) = \lim_{x \rightarrow 2^+} \left(\frac{x-2}{x^2-4} \right) = \lim_{x \rightarrow 2^+} \frac{\cancel{x}-2}{(\cancel{x}-2)(x+2)}$$

$$= \lim_{x \rightarrow 2^+} \frac{1}{x+2} \stackrel{\text{direct subst.}}{=} \frac{1}{2+2} = \left(\frac{1}{4} \right) \quad x=2$$

$$\begin{aligned} \lim_{x \rightarrow 2^-} \left(\frac{|x-2|}{x^2-4} \right) &= \lim_{x \rightarrow 2^-} \left(\frac{2-x}{x^2-4} \right) = \lim_{x \rightarrow 2^-} \frac{-(x-2)}{(x-2)(x+2)} \\ &= \lim_{x \rightarrow 2^-} \frac{-1}{x+2} \stackrel{\substack{\text{direct} \\ \text{subst.} \\ x=2}}{=} \frac{-1}{2+2} = -\frac{1}{4} \end{aligned}$$

one-sided limits are not equal $\Rightarrow \lim_{x \rightarrow 2} \left(\frac{|x-2|}{x^2-4} \right)$ DNE

to Summarize:

Limit technique #4: piecewise functions

if a function is defined in pieces, analyze both one-sided limits and compare.

Limits and inequalities

Theorem If $f(x) \leq g(x)$ when x is near a , then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$$

g is "squeezed" in between f and h

The squeeze Theorem If $f(x) \leq g(x) \leq h(x)$ when x is near a and if we also know

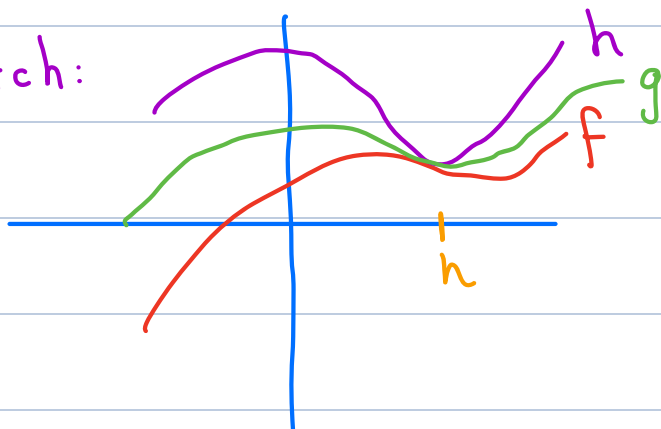
$$\lim_{x \rightarrow a} f(x) = L, \quad \lim_{x \rightarrow a} h(x) = L$$

then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x) \leq \lim_{x \rightarrow a} h(x)$$

$$\Downarrow$$
$$\boxed{\lim_{x \rightarrow a} g(x) = L}$$

Sketch:



Example: use the squeeze Theorem to evaluate

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$$

idea: $-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$

multiply everything by x^2 : $\underbrace{-x^2}_f \leq \underbrace{x^2 \cdot \sin\left(\frac{1}{x}\right)}_g \leq \underbrace{x^2}_h$

$$\lim_{x \rightarrow 0} -x^2 = \underset{\text{direct subst.}}{-0^2} = 0$$

$$\lim_{x \rightarrow 0} x^2 = \underset{\text{direct subst.}}{0^2} = 0$$

$\Rightarrow$ by squeeze Theorem,

$$\boxed{\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0}$$

to summarize:

Limit technique #5: squeeze theorem

if you can bound your function between two functions that approach the same limit, it will also approach that limit

Substitutions

Sometimes it's helpful to introduce a substitution to simplify an expression.

Example: a) evaluate $\lim_{x \rightarrow 1} \frac{3^{2x} - 9}{3^x - 3}$

b) evaluate $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x}$

solutions:

a) let $u = 3^x \leftarrow$ continuous function
rewrite limit in terms of u .

$$3^{2x} = (\underbrace{3^x}_u)^2 = u^2$$

$$\frac{3^{2x} - 9}{3^x - 3} = \frac{u^2 - 9}{u - 3}$$

$$\lim_{\boxed{x \rightarrow 1}} \frac{3^{2x} - 9}{3^x - 3} = \lim_{u \rightarrow 3} \frac{u^2 - 9}{u - 3} = \lim_{u \rightarrow 3} \frac{\cancel{(u-3)}(u+3)}{\cancel{u-3}}$$

$\uparrow$ as $x \rightarrow 1$, $u = 3^x \rightarrow 3$

$$\therefore \lim_{u \rightarrow 3} u + 3 = \boxed{6}$$

$$b) \lim_{x \rightarrow 0} \frac{\sin(3x)}{x}$$

$$u = 3x \Rightarrow x = \frac{u}{3}$$

$$\frac{\sin(3x)}{x} = \frac{\sin(u)}{u/3} = 3 \frac{\sin(u)}{u}$$

$$\lim_{x \rightarrow 0} \frac{\sin(3x)}{x} = \lim_{u \rightarrow 0} 3 \frac{\sin(u)}{u}$$

↑
as $x \rightarrow 0$, $u = 3x \rightarrow 0$

$$\begin{aligned} &\xrightarrow{\text{by limit laws}} 3 \lim_{u \rightarrow 0} \frac{\sin(u)}{u} = 3 \cdot 1 = \boxed{3} \end{aligned}$$

to summarize:

Limit technique #6: substitutions

introduce a substitution u = some function of x

rewrite limit as a limit in terms of u